x and y are two $\overset{m \times k}{\wedge}$ 2 way tables which have same row totals & column totals.

Following the algorithm outlined in class

$\begin{matrix} + & -1 \\ - & +1 \end{matrix}$ the chain can move from x to y.

$$x = \begin{array}{c|c} \begin{bmatrix} x_{11} & x_{12} & & x_{1k} \\ x_{21} & & & \\ \vdots & & & \\ x_{m1} & & & x_{mk} \end{bmatrix} & \begin{matrix} x_{1\cdot} \\ \\ \\ x_{m\cdot} \end{matrix} \\ \hline \begin{matrix} x_{\cdot 1} & x_{\cdot 2} & & x_{\cdot k} \end{matrix} & N \end{array}$$

$x_{i\cdot} = y_{i\cdot} \quad \forall i$

$x_{\cdot j} = y_{\cdot j} \quad \forall j$

Suppose $m = 2$.

$$\begin{bmatrix} x_{11} & x_{12} & & x_{1k} \\ x_{21} & x_{22} & & x_{2k} \end{bmatrix} \longrightarrow \begin{bmatrix} y_{11} & \cdots & y_{1k} \\ y_{21} & \cdots & y_{2k} \end{bmatrix}$$

column 1 using column 2

If $y_{11} = x_{11}$ then move to next step

$y_{11} > x_{11}$ $\quad i=1,\ i'=2,\ j=1,\ j'=2$.

$x_{11} \to x_{11}+1$

$x_{12} \to x_{12}-1$

$x_{21} \to x_{21}-1$

$x_{22} \to x_{22}+1$

check if $y_{11} > x_{11}+1$.
Then do same operation again
Stop when $y_{11} = x_{11}+\ldots$

If $y_{11} < x_{11}$ $\quad i'=1,\ i=2,\ j=1,\ j'=2$

$x_{11} \to x_{11}-1$

$x_{12} \to x_{12}+1$

$x_{21} \to x_{21}+1$

$x_{22} \to x_{22}-1$

Continue until $y_{11} = x_{11} - \ldots$

Once x_{11} and y_{11} are same, x_{21} will also be same as y_{21}

Match column 2 using column 3

$\vdots$

Once first $\overset{k-1}{(k-1)}$ columns match, the $\overset{k}{k}$-th column also has to match.

We have shown that for two $\boxed{2} \times k$ tables x ~~any~~ and y with same row & column total the chain can move from x to y in finite number of steps following the $\begin{pmatrix} + & - \\ - & + \end{pmatrix}$ algorithm

Suppose for two $m \times k$ tables w. . . . . . for all $m = 2, 3, \ldots, n$.

Consider x & y two ~~n+1~~ $(n+1) \times k$ tables with same row & column totals

$$x = \begin{bmatrix} x_{11} & \circ & \_ & x_{1k} \\ \circ & \circ & & \\ \vdots & & & \\ x_{(n+1)1} & & & x_{n+1\,k} \end{bmatrix} \qquad y =$$

Using x_{21}, x_{22}, x_{12} you change x_{11} to y_{11}.

Continue to match x_{1i} to y_{1i} for $i = 2, \ldots, k-1$.

First rows of x and y are matched.

The remaining rows form two $n \times k$ tables x' and y'

The row totals are same.

The column totals are the difference between the original column totals & the first ~~col~~ row. So these are also same

Improved Simulation

Pick $i, i' \in \{1, \ldots m\}$, $i \neq i'$ $\quad j, j' \in \{1, \ldots, k\}$, $j \neq j'$.

$x_{ij} \to x_{ij} + e$

$x_{ij'} \to x_{ij'} - e$

$x_{i'j} \to x_{i'j} - e$

$x_{i'j'} \to x_{i'j'} + e$

$$\begin{array}{cc|c} \begin{bmatrix} x_{ij} & x_{ij'} \\ x_{i'j} & x_{i'j'} \end{bmatrix} & & \begin{matrix} c \\ d \end{matrix} \\ \begin{matrix} a & \quad b \end{matrix} & & \end{array}$$

Consider all 2×2 tables with row & column sums a, b, c, d and pick one at random.

Midterm Part-B

① (a) **Null** :- $F \equiv G$ (deaf; Hearing) $\rightarrow$ ③

Alternate :- F is shifted right to G — ②

$X_1, X_2, X_3, \ldots, X_{10} \sim F$

$Y_1, Y_2, Y_3, \ldots, Y_{10} \sim G$ — ②

(b) — ③

(c) $\sum R_{x_i} = 150.5 \rightarrow$ ⑤

$T = 95.5 \rightarrow$ ③ — 6

(d) $m = n_1 + n_2 = 10 + 10 = 20$ — ①

$n = 10$ — ①

$T = 95.5$ — ② — ①

② (a) $P(X < \mu - a) = P(X > \mu + a) \ \forall a \rightarrow$ ④

(b) $F_{X-\mu}(a) = P(X - \mu \leq a)$

$= P(X \leq a + \mu)$

$= 1 - P(X \geq a + \mu)$

$F_{\mu - X}(a) = P(\mu - X \leq a)$

$= P(X \geq \mu - a)$

$= 1 - P(X \leq \mu - a)$

Since $1 - P(X \geq \mu + a) = P(X \leq \mu - a)$

$\Rightarrow F_{X-\mu}(a) = F_{\mu - X}(a) \rightarrow$ ⑥

③ (a) $\hat{f}(7) = \dfrac{\sum w_i(x) y_i}{\sum w_i} =$ — ①

$w_i = K_2(x_i - 7)$ — ①

$w_i = 0$ except for $x_i = 6, 8$

$w_i = \frac{3}{8}\left(1 - \frac{1}{4}\right) = \frac{3}{8} \times \frac{3}{4} = \frac{9}{32}$ — ④

$$\hat{f}(7) = \frac{\frac{9}{32} \times 1.05 + \frac{9}{32} \times (-1.05) + \frac{9}{32} \times (-1.37)}{\frac{9}{32} + \frac{9}{32} + \frac{9}{32}}$$

$= -0.4633$ — ④

(b) $f(7) = 7 - \frac{7^2}{10} = 2.1$ — ③

Error $= -0.4633 - 2.1 = -2.5633 \rightarrow$ ②

(c) $K_3(t) = \frac{1}{\cancel{34}}\left(1 - \frac{t^2}{9}\right) \rightarrow$ ⑤